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UX

Deep Dive into the UX Field

An interview with an Innovation Specialist at U.S. Department of the Interior's Office of Natural Resources Revenue (ONRR).



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User Experience (UX) design involves understanding what a person might think and feel when interacting with a product. As more organizations see the value of the field, there's a growing number of uses for UX design as well as a growing alphabet of UX-related careers – from information architect to visual designer. To help you keep up to date with the terms, I am sharing some insight into the field by speaking with UX experts.

For our first deep dive, I spoke with Shannon McHarg, an Innovation Specialist at the U.S. Department of the Interior's Office of Natural Resources Revenue (ONRR), which is responsible for collecting payments on oil, gas, and mineral extraction on federal and Native American lands and waters. Her team handles the site, revenuedata.doi.gov and (in the near future) ONRR.gov. *



Stephanie: *What does an Innovation Specialist do at ONRR?*

Shannon: We've got three Innovation Specialists [at ONRR], and we all have a different focus. I'm UX; then we have a content strategist and a developer. Our

website was started by [18F](#) (an office that works to improve technology use within the federal government), who initially created [revenuedata.doi.gov](#), and the Innovation Specialists were hired on by ONRR with recruitment support from 18F as part of the transition from 18F to an internal team. The hiring model had the “Innovation Specialist” job title partly because it was the easiest way to hire highly skilled people. But we are term-limited, meaning it’s a two-year role plus the potential for a two-year extension. I focus on UX but we’re also focusing on training career government employees to do the work that we do, so the website can continue after we term out. Day to day, I do the UX work of research with users, I design, and then I also train program analysts to do the user research and design. For example, we did a mock study where they did all the parts of the user research process with people on our team pretending to be users, and now they’re starting to facilitate some of sessions with real users and analyze findings. So, they’re picking up pieces bit by bit along the way.

Stephanie: *Who do you work with on a regular basis to complete projects?*

Shannon: We have an agile team that works on [revenuedata.doi.gov](#), an open-data website, which shows all of the money going in and out of our office. That team is made up of a product manager, the three innovation specialists, and three program analysts who do a lot of data analysis and are trying to learn digital skills. We also have two contract developers because we needed additional development capacity and two Virtual Student Federal Service interns who are learning UX. We’ve also got a sister team within the agency who does all of the retrieval of our data, so they give us Excel files that we use to put on the website.

Stephanie: *How did you get to become an Innovation Specialist?*

Shannon: I have been doing UX for about 17 years. I have a master’s degree from Bentley University in Human Factors in Information Design. So, I kind of took a non-typical path of going to school for it and coming out of school working in the field. Most people do not do that in UX. I worked in the private sector up until this position. I worked at H & R Block for almost a decade, which is very similar to the government in a lot of ways where they were built up as a not digital company but had to transition into having digital products. Then I worked at L.L. Bean for four years, which was built up around catalog and had to become digital. Afterwards, I worked at a small agency that worked with a variety of clients from startups to a bunch of other large companies. I then decided I wanted to work in government and started applying to government UX positions, and eventually landed the Innovation Specialist position with ONRR.

Stephanie: *Any key takeaways from those experiences that might be applicable to the federal work environment?*

Shannon: [Regarding school experiences relevant to skills for working in the government], I think I learned most of it on the job. Skills related to organizational dynamics and changing those dynamics are the type of things you don't realize until you actually get into the situation in the real world. My first boss was a great mentor. He was able to help me figure out how to observe and adapt to the environment I was working in – things like being able to pick up cues from people who have been in the situation for a long time and observing meetings to figure out what's going on and how can we make progress. For example, when I started working at H & R block, we did wireframes in Microsoft Word because that was the easiest way for developers to copy and paste the content. Being able to be flexible with whatever tools you have available, not get so hung up on things like not having Sketch, and being able to work within whatever process currently exists make it easier to get small inroads that can go a long way to changing an organization over time.

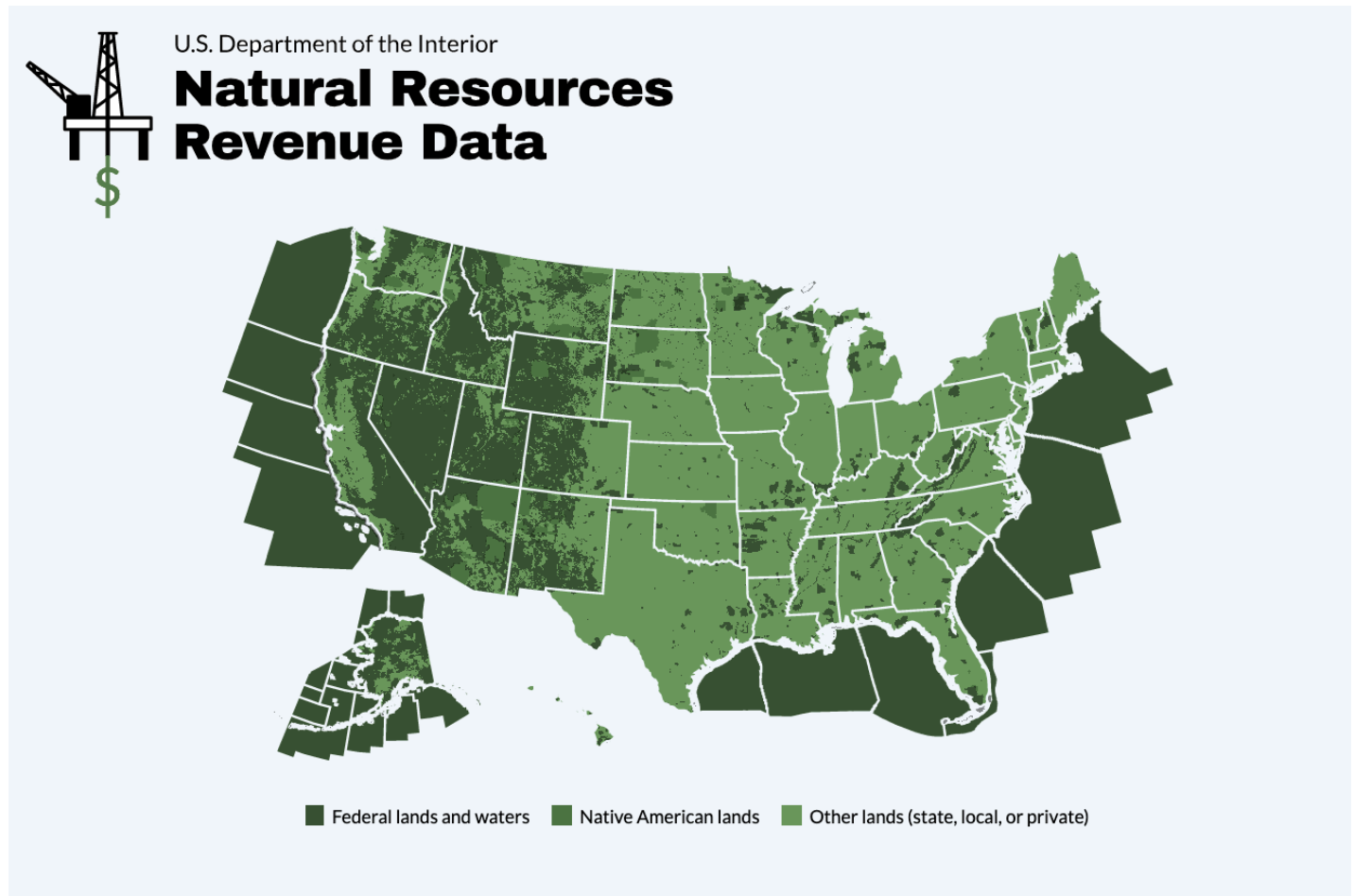
Stephanie: *As an Innovation Specialist, what are some of the most challenging parts of your job?*

Shannon: Organizational inertia and silos. For example, it's hard for us to get information about some of our data because we don't always know who the right expert to contact is. We have to measure some obscure minerals that are mined on federal land that utilize unique measurement units that we needed information about to define for users. When digging around, we couldn't find anyone who even knew what [the unit] was. Just trying to figure out who to talk to for things is fairly challenging. And also, because everyone has their own area of expertise, there's not a lot of sharing across areas. But it's definitely a fun challenge to figure out – like, now that we've found the person who knows about quartz crystal, it's now a challenge to figure out how to get them talking to people in other silos within the organization about the things they know. It's also challenging to figure out how to make websites sustainable once we term out. Our program analysts probably won't have the full skillset to build a website by the time we term out, so it's challenging to prioritize what to teach them and figure out what alternatives might be to get the work done after we leave.

Stephanie: *What's your ideal vision for making the website sustainable?*

Shannon: We're trying to do things to make it possible for data to be updated by uploading a spreadsheet. It used to take a developer and a development machine to be able to run the code to update the data. We changed the code

base to simplify that backend process. In order to do that, we have had to postpone a lot of the work to make it easier for users [on the frontend] in order to make it easier for the people on the backend to update the site. So, it affects the prioritization of the website.



Dark green areas of the map indicate federal lands and waters, a medium shade of green shows Native American lands, and light green is used on other lands (state, local, or private).

Stephanie: *What are some of the most exciting parts of your job?*

Shannon: It's always fun to watch people use the website. It often seems so obvious once you see users encounter issues in testing, but since you're so close to working with the site you often don't see the thing that's right in front of you until someone with fresh eyes encounters it. It's a lot of fun to see that; you can then make a small change and make things so much easier for the user.

It's also a lot of fun being able to bring UX to an organization that hasn't done a lot of it. Because we're internal, I'm able to sit on a strategic planning team right now, and I'm working with people in the organization that I never worked with before. One of the things we're looking at is how we deal with stakeholders external to our agency, and that's just something they've never done before.

Being able to talk with people from across the agency about external stakeholders and seeing them talking with people outside the organization about their experience is really exciting because no one has ever talked to many of them about their needs before. Being able to spread that feedback loop, and the type of thinking that goes with that, is really exciting to see, and it'll be fun to see how things evolve over time.

Stephanie: *Looking out to the future (or in the present), how do you see the field of UX careers evolving, specifically within government?*

Shannon: It'll be interesting to see how it plays out because every agency, I think, is doing it differently right now. Whether it ends up being teams like 18F or the [U.S. Digital Service](#) (USDS) driving a lot of the inroads into it, or if it ends up being more homegrown with people who are already in agencies picking up the skills — or some mix of that, like our positions — it will be interesting to watch. It'll be interesting to see what sticks and can be carried over to other agencies.

Stephanie: *Any other agencies whose UX work we can look to as an example that you'd recommend?*

Shannon: I haven't seen a lot of them presenting or publishing information. But the [Lab at OPM](#) seems like they are doing really interesting work. I wish more groups would publish what they're doing so that we can learn from it. We have a [blog](#) where we're trying to be completely open about everything we're doing. It would be great to see others doing that as well so we can all learn from each other.

Stephanie: *What kind of tools do you recommend?*

Shannon: Tools change so often. All you need is something you can do a wireframe in, something you can do remote user sessions with, with video, something that you can track issues with, and something you can use to communicate among your teammates. We use Axure (tool to make clickable prototypes) just because we were able to get approval for it. And our agency just switched from Google to Microsoft so we're changing all of our tools. You just have to be flexible in working with whatever tools you have. The skills to focus on users are much more important than the skills in any one tool.

Stephanie: *What tips might you have for an office within the government that is looking to add on new UX roles to their workforce?*

Shannon: It makes sense to assess what they have in-house already. So, if they have someone interested in doing UX, they can invest in training that person. Or

if they're looking to hire, trying to find the most skilled people they can. I recommend looking at what hiring authorities are available within their agency and any other ways available to get highly skilled people. Dealing with regulations and organizational dynamics within government is really hard, so it's better to have someone who has the UX skills down cold so they can focus on using those skills to help drive organizational change and training people within the agency in the UX skills. It will get you a lot further than hiring someone junior.

** Note: This interview has been edited for clarity and brevity.*

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[Stephanie Santana](#)

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